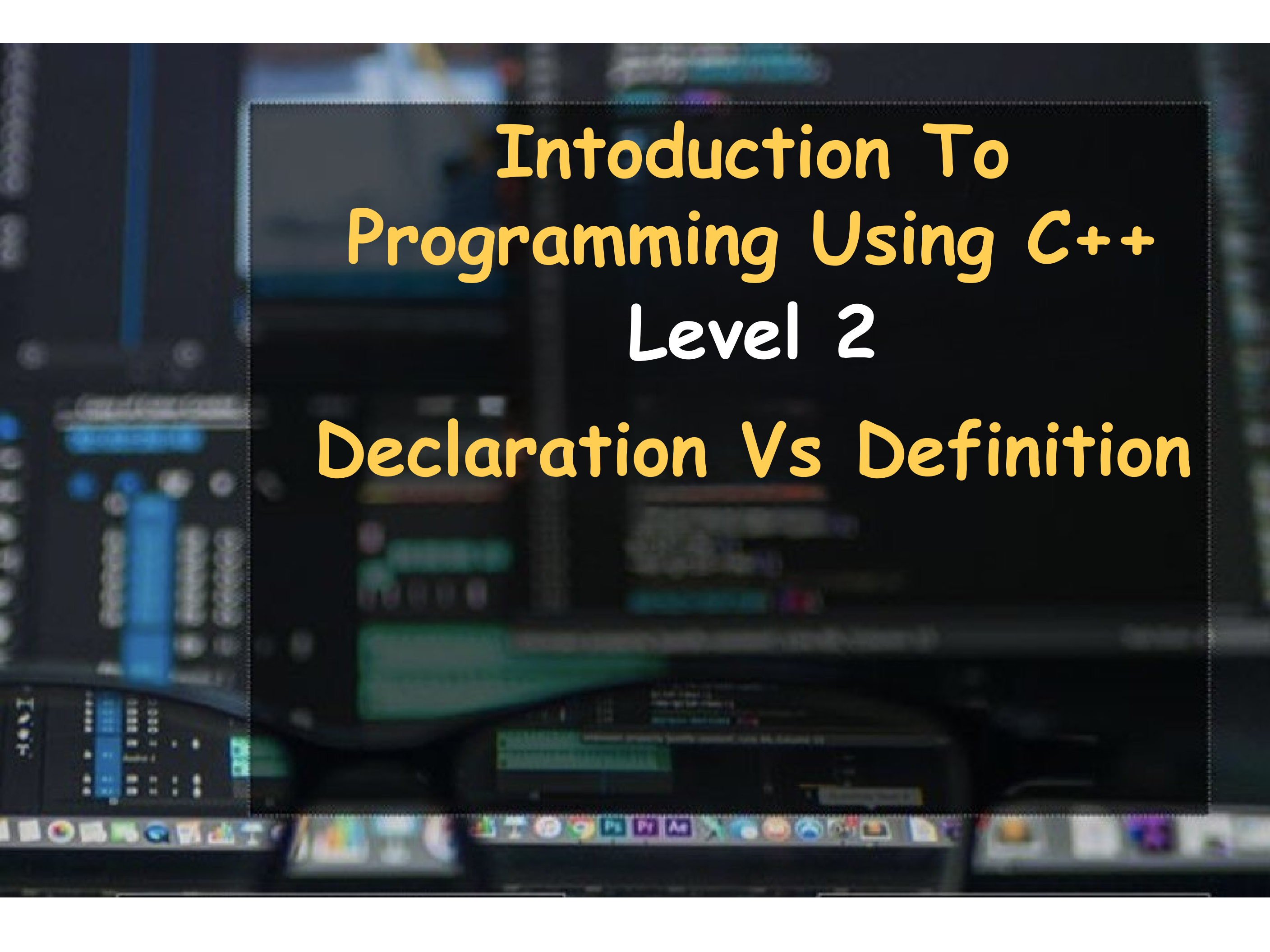
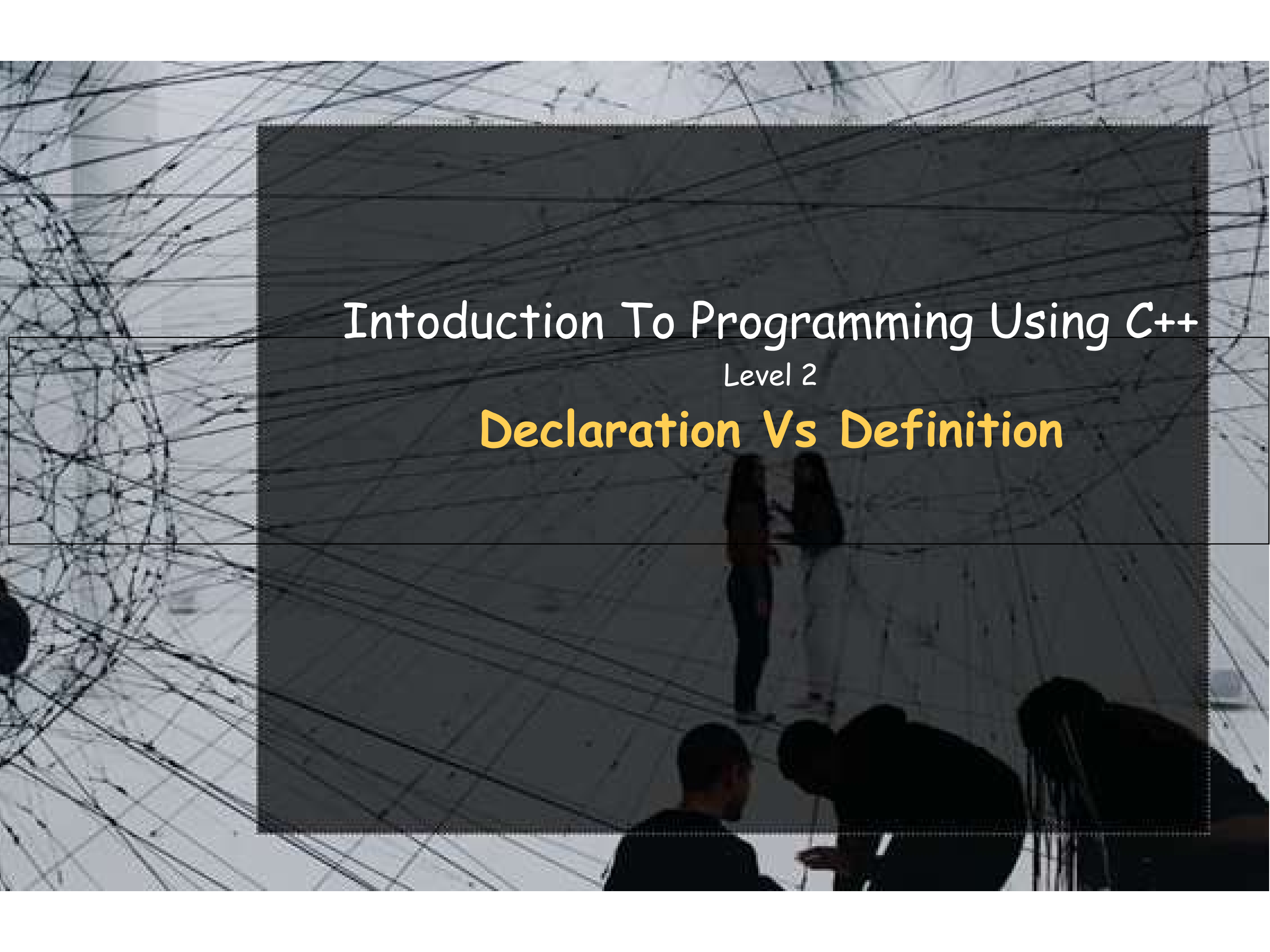




Introduction To Programming Using C++ Level 2 Declaration Vs Definition



Introduction To Programming Using C++

Level 2

Declaration Vs Definition

Definition

■ A Function Declaration tells the compiler that a function exists and provides its name, return type, and parameters.

■ A Function Definition contains the actual implementation, including the function body.

▨ If you want to call a function before its Definition, you must declare it before ``main()`` so the compiler knows about it.

▨ If the function Definition is written after ``main()``, a Declaration before ``main()`` is needed.

Explanation

Declaration:

```
void add(int, int);
```

 This tells the compiler that a function named `add` exists, returns `void`, and takes two `int` parameters.

Definition:

```
void add(int a, int b)
{
    cout << (a + b);
}
```

 This contains the actual code that the function executes.

Procedures

1. Write the Function Declaration before ``main()`` if the function Definition will be written later.
2. Call the function inside ``main()``.
3. Write the Function Definition after ``main()``.

💡 Example

```
void add(int, int);    // Declaration
int main()
{
    add(10, 20);
    return 0;
}
void add(int a, int b) // Definition
{
    cout << (a + b);
}
```

- ⚡ The compiler sees the Declaration before `main()`, so it knows that `add()` exists.
- ⚡ The actual implementation of `add()` is provided later in its Definition.

➡ Result

The function is called with:

10 + 20

The output is:

30

Note

- ◆ A Declaration tells the compiler "this function exists."
- ◆ A Definition tells the compiler "this is what the function does."
- ◆ If the Function Definition is already written before `main()`, a separate Declaration is not required.

⊘ Warning

- ◆ The Declaration and Definition must match in the function name, return type, and parameter types.
- ◆ A Declaration alone does not provide the function's implementation.



Comparison

▨ Declaration

- ✓ Tells the compiler that the function exists.
- ✓ Usually written before `main()` when the Definition comes later.
- ✓ Does not contain the function body.

▤ Definition

- ✓ Contains the actual implementation of the function.
- ✓ Contains the function body `{ }`.
- ✓ Specifies what the function actually does.

✦✦ Summary

- ✓ Declaration tells the compiler about the function.
- ✓ Definition contains the implementation.
- ✓ A function can be declared before ``main()`` and defined after ``main()``.

Thank you 😊